



OLLSCOIL NA GAILLIMHE
UNIVERSITY OF GALWAY

Semester 2 Examinations 2025/2026

Course Instance Code(s) 4BCT, 3BP, 4BP, 3BLE, 4BLE, 1OA, 1EM
Exam(s) B.Sc. (Computer Science & Information Technology),
B.E.(Electronic and Computer Engineering), B.E.
(Electrical & Electronic Engineering)

Module Code(s) CT420
Module(s) Real-Time Systems
Bonded No ☒] Yes []
Paper No. 1

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Instructions: This exam has two sections:
In section A, answer any *two* questions between Question 1, 2,
and 3.
In section B, answer any *two* questions between Question 4, 5
and 6.

Duration 2 hours
No. of Pages 6
Discipline(s) Computer Science

Requirements:

Take from exam venue, release online	No []	Yes [x]
MCQ Answer sheet	No [x]	Yes []
Log Tables	No [x]	Yes []
Cambridge Tables 2 nd Edition	No [x]	Yes []
Graph Paper	No [x]	Yes []
Graphic material in colour	No []	Yes [x]

End of requirements.

Section A

Answer any two questions out of three in Section A

Question 1 (15 Marks)

- (a) The future exploration and colonisation of the moon will require robust localisation for navigation on the moon's surface. Discuss a technology stack and its inner workings that can be used to provide lunar surface navigation. Your proposal can be based on an existing (terrestrial) technology.

[5 marks]

- (b) Hardware timestamping, transparent clocks, as well as the P2P delay calculation mechanism are some of the outstanding features of the PTP time synchronisation protocol. Using an example (that entails a network diagram, as well as all relevant PTP messages and timestamps exchanged), show how all these mechanisms work hand-in-hand to provide the accurate offset calculation of a slave clock against a grandmaster clock.

[6 marks]

- (c) An alternative way to correct computer clocks is by using an internal clock synchronisation algorithm like the Berkeley Algorithm. Using an example, show how this method works, and how local clocks that are completely out of sync are dealt with.

[4 marks]

PTO

Question 2 (15 Marks)

- (a) You are a senior system development engineer for an automotive company and are tasked to select an RTOS to be used in the company's latest product line of navigation systems. Outline 4 key areas you would deem necessary for this RTOS to support real-time applications. These key areas can be based on the POSIX API. Use examples to support your answer.

[6 marks]

- (b) Explain the term Hamming Distance (HD). Determine the HD of the following:
- Hamming (7, 4) code
 - 8-bit ASCII with parity bit
- Fully explain your answers.

[4 marks]

- (c) Do an RM schedulability analysis on the set of 4 tasks shown below, focussing on W4 and its associated checkpoints. Subsequently, determine the RM task schedule over the interval [0, 400]:

i	e_i	p_i
1	20	100
2	20	180
3	80	240
4	100	400

[5 marks]

PTO

Question 3 (15 Marks)

- (a) Using (pseudo) C-code develop a cyclic executive scheduler for the set of tasks below. Your code must be able to detect and count the number of task overruns. Fully explain your answer.

Task	Period p [ms]	Exec Time [ms]
A	25	10
B	25	8
C	50	5
D	50	4
E	100	2

[5 marks]

- (b) Discuss how access to a critical region can be controlled via a mutex (binary semaphore). Using (pseudo) code show how a mutex can be implemented in software.

[5 marks]

- (c) Memory locking is an important feature of POSIX.4. Use the 7-state process model in conjunction with an example written in C to outline how memory locking works. Use pseudo code if you can't recall the exact POSIX API calls and explain what would happen if memory locking (i) was not used and (ii) too generously used.

[5 marks]

PTO

Section B

Answer any two questions out three in Section B

Question 4 (15 Marks)

- (a) Explain the concept of Head-of-Line (HoL) blocking at the transport layer (TCP). Describe how QUIC eliminates transport-layer HoL blocking and why this is particularly important for soft real-time systems.

[7 Marks]

- (b) Adaptive bitrate (ABR) streaming is widely used in video delivery systems.

Explain:

1. How ABR works
2. The role of bandwidth estimation
3. The trade-offs between video quality and latency
4. Discuss why ABR alone is insufficient for highly interactive real-time applications.

[8 Marks]

Question 5 (15 Marks)

- (a) Briefly explain what type of information qLog captures in a QUIC connection and how qViz helps identify issues in QUIC protocol.

[4 Marks]

- (b) Explain the concept of fairness in congestion control.

Compare:

1. Throughput fairness
2. Latency fairness
3. Discuss why fairness becomes more complex in mixed TCP and QUIC environments.

[5 Marks]

- (c) Discuss the role of encryption in QUIC.

Explain:

1. Why QUIC encrypts most transport metadata
2. The advantages of this design
3. The challenges it creates for network monitoring and middleboxes

[6 Marks]

PTO

Question 6 (15 Marks)

Case Study: FastPlay Cloud Gaming Platform

FastPlay is launching a global cloud gaming service. The system architecture is as follows:

- Game engines run in regional cloud data centres.
- Video is encoded in real time (H.265).
- Streams are delivered to clients using TCP.
- Users connect from home broadband and mobile networks.
- The company reports the following issues during beta testing:
 1. Users experience noticeable input lag during peak hours.
 2. Mobile users experience temporary freezes when switching between WiFi and 5G.
 3. Some users report fluctuating video quality.
 4. Network logs show increased packet loss and variable RTT during congestion.

- (a) Identify and explain the main sources of end-to-end delay in this system. Break down the delay components and explain which ones are most critical for interactive cloud gaming.

[5 Marks]

- (b) Explain how QUIC features could help mitigate issues (2) and (4). Discuss specifically:
1. Stream multiplexing
 2. Connection migration
 3. Congestion control
 4. Loss recovery

[10 Marks]

END OF EXAM